Homework 2 - 09/10 Content

MATH 1271, Garrett Credi

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Problems listed as AC are indexed as [Chapter].[Section].[Problem Number] and are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are indexed as [Chapter].[Section].[Problem Number] and are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic].[Section].[Problem Number]

1 Type 1 - Limits Graphically

Choose 4 of the following problems to solve:

1. PAULS-PRACT.Limits.The Limit.4-6 (each of these counts as one problem)
2. PAULS-ASSIGN.Limits.The Limit.5-7 (each of these counts as one problem)
3. AC 1.2.1

2 Type 2 - Estimating Limits

Choose 2 of the following problems to solve:

1. PAULS-ASSIGN.Limits.The Limit.1-4 (each of these counts as one problem)
2. AC 1.2.6
3. APEX 1.1.21-28 (each of these counts as one problem)

3 Type 3 - Algebraically Computing Limits

Choose 4 of the following problems to solve:

1. PAULS-ASSIGN.Limits.Computing Limits.5-20 (each of these counts as one problem)
2. AC.1.1.3-5 (Each of these counts as one problem)

4 Challenge Problem

Suppose that you drop a ball and know that it's height in meters as a function of time (in seconds since you dropped it) by:

$$h(t) = \begin{cases} 40 - 10t^2, & 0 \leq t \leq 2 \\ -10t^2 + 60t - 80, & 2 \leq t \leq 4 \\ -10t^2 + 90t - 200, & 4 \leq t \leq 5 \\ 0, & \textit{otherwise} \end{cases} \tag{1}$$

1. What is $\lim_{t \to 1} h(t)$?
2. What is $\lim_{t \to 2} h(t)$?
3. What is the instantaneous velocity of the ball at $t = 1$?
4. What is the instantaneous velocity of the ball at $t = 2$?